HEALTHCARE PATIENT RECORD ANALYSIS

Submitted in partial fulfillment of the requirements

for the Vocational Course (BCA-261)
Batch (2024-2025)

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DECLARATION

I hereby declare that the Project titled **HEALTHCARE PATIENT RECORD ANALYSIS** submitted partial fulfillment of the Vocational Course **BCA 261 – Advanced Excel and Power BI** is my original work and has not been plagiarized or reproduced from any other source.

This project has been completed under the guidance of **Ms. Suman Singh** and is in partial fulfillment of the requirements of the course conducted by **Jagan Institute of Management Studies**.

All sources of information used have been acknowledged appropriately, and I understand that any violation of academic integrity will result in appropriate disciplinary action.

Date of Submission: 25/08/2025

Place: Jims, Rohini sec-5, Delhi - 110085

Signature: _____

Name: **DIVYANKA CHANDHOK**

Student Enrollment Number: **02650402024**

CERTIFICATE

This is to certify that **DIVYANKA CHANDHOK** has successfully completed the project titled "Healthcare Patient Record Analysis and Visualization using Tableau & Power BI" conducted as part of the academic curriculum for the academic year 2024–25.

We commend her dedication, sincerity, and effort in acquiring practical knowledge and hands-on skills in the field of Data Analytics and Business Intelligence. This achievement reflects her ability to apply theoretical concepts to real-world scenarios with commitment and excellence

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Signature:

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to all those who supported and guided me throughout the completion of this **HEALTHCARE PATIENT RECORD ANALYSIS** undertaken as part of the Vocational Course BCA 261 - **Advanced Excel and Power BI.**

First and foremost, I extend my heartfelt thanks to **Ms. Suman Singh**, whose valuable guidance, encouragement, and insights were instrumental in shaping this project.

I also wish to thank the faculty and staff of **Jagan Institute of Management Studies** for creating a learning environment that fostered curiosity and innovation.

My appreciation also goes to my peers and friends who shared their feedback and offered support during the course.

Finally, I acknowledge the contributions of all data sources, technical platforms, and tools used in the research and execution of this project.

Signature: _____

Name: **DIVYANKA CHANDHOK**

Student Enrollment Number: **02650402024**

List of Tables/Figures

LIST OF TABLES

Table No	Description	Page No
3.1.1	Dataset of HEATHCARE PATIENT RECORD ANALYSIS	10-14
3.2.5	Measure 1: Total Sales	19
3.2.6	Measure 2: Average Discount	19
3.2.7	Measure 3: Return Rate	20
3.2.8	Measure 4: Profit	20

LIST OF FIGURES

Figure No	Description	Page No
3.2.1	Bar chart: Sum of quantity sold by region	15
3.2.2	Line chart: Sales trend over time	16
3.2.3	Pie chart: Sale distribution by category	17
3.2.4	Scatter chart: Unit Price VS Quantity	18
4.1.1	Tableau dashboard for measures and charts	21

TABLE OF CONTENTS

S.No.	Topic	Page No.
1	Declaration	1
2	Certificate	2
3	Acknowledgements	3
4	List of Tables/Figures	4
5	**1. - Introduction** **1.1 -** Background of the Topic **1.2 -** Objective and Scope of the Project **1.3 -** Relevance to the Project	7-8
6	**2. - Methodology** **2.1 -** Data Source **2.2 -** Tools and Techniques	9
7	**3. - Data Collection & Analysis** **3.1 -** Presentation of Data **3.2 -** Charts, Tables, and Interpretation	10-20
8	**4. - Findings and Observations** **4.1** – For Tableau - Dashboard with tables and Charts, Outcomes For Power Bi- **4.2** - Dashboard with Tables, Outcomes **4.3** Dashboard with Charts, Outcomes	21-22

9	**5. - Conclusion and Recommendations** **5.1** - Summary of Learnings **5.2** - Suggestions for Future Work	23-24
10	References	25
11	**Annexures** **A.** - Raw Data / Questionnaire **B.** - Screenshots of Work **C.** - Additional Tables or Supplementary Material	24

Introduction: Healthcare Patient Record Analysis

The healthcare industry generates massive volumes of patient data every day, ranging from clinical records and treatment histories to billing and demographic information. Managing and analyzing this data is essential for enhancing patient care, streamlining operations, and making informed decisions.

This project, **"Healthcare Patient Record Analysis,"** aims to explore and visualize key insights from patient records using **Tableau**, a powerful data visualization tool. The focus is on identifying trends in patient visits, regional healthcare performance, treatment categories, and financial metrics such as total sales, discounts, and profit margins.

By analyzing patient records, this project seeks to:

- Understand patterns in patient behavior and treatment frequency
- Highlight performance differences across regions
- Track healthcare sales and return rates
- Examine the impact of pricing and discounting strategies
- Support data-driven decision-making in healthcare management

1.1 BACKGROUND OF HEALTHCARE

Healthcare is a critical sector that encompasses the prevention, diagnosis, treatment, and management of illness, as well as the preservation of mental and physical well-being through medical services. It involves a wide network of professionals, institutions, technologies, and systems that work together to provide patient-centered care.

In recent decades, the healthcare industry has seen rapid growth due to advancements in medical technology, increased awareness of public health, and rising global health challenges. This growth has

also led to the **generation of vast amounts of data** — from patient medical records and prescriptions to insurance claims and operational performance metrics.

Efficiently managing and analyzing this data is vital for:

- Enhancing the quality of care
- Reducing operational costs
- Improving patient outcomes
- Identifying public health trends
- Supporting policy-making and strategic decisions

1.2 – Objective and Scope of the Project

The primary objective of this project is to analyze visualize healthcare patient records using **Tableau** to uncover meaningful insights that can improve decision-making in the healthcare sector. Through data visualization, the project aims to:

- Identify trends in patient visits, treatments, and sales over time
- Evaluate regional performance in terms of healthcare delivery and patient volume
- Analyze key financial metrics such as total sales, discounts, return rates, and profits
- Understand category-wise treatment/product usage and demand
- Enable interactive exploration of patient data for better strategic planning

This project focuses on the analysis of **structured patient and sales data** within a healthcare setting. The scope includes:

- Importing and cleaning patient-related datasets
- Creating interactive **Tableau dashboards** to visualize KPIs and trends
- Comparing performance across different **regions, categories, and time periods**
- Exploring relationships between variables such as unit price, quantity, and profit
- Presenting findings in a visually appealing and easily interpretable format

1.3 - Relevance to the project

The healthcare industry is increasingly dependent on **data-driven insights** to enhance patient care, optimize operations, and improve financial performance. With the exponential growth of patient data, it has become essential to implement tools and techniques that can transform this raw information into **actionable insights**.

This project is highly relevant in today's context, as it demonstrates how **data visualization** through **Tableau** can play a crucial role in:

- **Improving decision-making:** By visualizing trends in patient visits, treatment types, and regional performance, healthcare providers can make informed strategic decisions.
- **Enhancing operational efficiency:** Identifying high-return areas, tracking discounts, and analyzing return rates helps in optimizing healthcare operations.
- **Monitoring performance:** The use of KPIs (Key Performance Indicators) enables organizations to track sales, profit, and service outcomes in real time.

- **Resource allocation:** Understanding category-wise demand and regional trends supports better allocation of medical resources and personnel.

This project bridges the gap between raw patient data and meaningful information. It highlights the power of Tableau as a business intelligence tool in healthcare analytics and underlines the importance of **visual storytelling** in complex data environments.

2. METHODOLOGY

2.1 – Data source

The dataset is given by the instructor for this project in the excel format, which contains sales data for HEALTHCARE PATIENT RECORD ANALYSIS , including fields such as ID, Date, Customer, Region, Category, Product, Quantity, Unit Price, Discount, and Returned.

2.2 - Tools and Techniques

1. **Tableau Desktop**
   - A powerful data visualization tool used to create interactive dashboards and perform in-depth analysis.
   - Used for connecting to datasets, cleaning data, and building charts, graphs, and visual stories.
2. **Microsoft Excel**
   - Used for initial data cleaning, formatting, and preprocessing before importing into Tableau.
   - Also used to maintain source files for structured patient records and sales information.

Techniques Applied:

1. **Data Cleaning and Preparation**
   - Removing null values, correcting data types, and formatting fields for consistency.
2. **Data Visualization**
   - Creating various visual elements such as:
     - Bar Charts (for regional sales comparison)
     - Line Charts (for monthly trends)
     - Pie Charts (for category-wise distribution)
     - Scatter Plots (for unit price vs. quantity analysis)
     - KPI Cards (for key metrics like profit, return rate, etc.)
3. **Dashboard Design**
   - Building interactive dashboards with filters, legends, tooltips, and dynamic visuals to allow end users to explore the data intuitively.
4. **Calculated Fields**
   - Used to derive additional insights such as profit margin, discount percentage, and average sales per region.

5. **Trend and Comparative Analysis**
   - Analyzing changes in metrics over time, and comparing performance across categories, products, and regions.

3. DATA COLLECTION & ANALYSIS

The data collection phase involved gathering relevant patient records from [specify source, e.g., hospital database, electronic health record (EHR) system, clinical data repository]. The data was collected in compliance with ethical standards and patient confidentiality protocols, such as HIPAA (Health Insurance Portability and Accountability Act) regulations.

3.1-Presentation of data

ID	Date	Customer	Region	Category	Product	Quantity	Unit Price	Discount	Returned
1	2023-04-04	Roberta Ryan	West	B	describe	11	493.34	0.03	Yes
2	2024-12-15	Jennifer Foster	South	A	huge	7	229.95	0.27	No
3	2021-07-21	Eric Carter	North	A	but	14	216.94	0.22	Yes
4	2021-11-09	Terry Lindsey	South	A	your	1	198.11	0.13	No
5	2021-06-03	Alicia Baker	North	C	fact	3	284.62	0.09	Yes
6	2021-08-26	Melissa Barnes	North	C	case	10	382.99	0.14	No
7	2021-06-10	Susan Adams	North	A	cut	2	197.22	0.21	Yes
8	2020-10-12	Philip Flores	West	B	main	1	454.53	0.17	No
9	2021-04-15	Jennifer Smith	36	C	through	14	103.7	0.09	No
10	2022-05-30	Barry Glover	East	B	throughout	11	407.83	0.05	Yes
11	2020-02-09	Kendra Lam	North	C	project	7	497.84	0.22	No
12	2023-10-03	Jeremy Thompson	East	B	garden	10	265.2	0.12	No
13	2023-12-24	Roger Williams	West	C	close	12	105.68	0.26	No
14	2025-05-27	Kristin Haynes	East	A	special	7	52.8	0.03	Yes

15	2020-12-25	Terry Leon	South	B	protect	15	345.31	0.08	No
16	2020-03-28	Clayton Taylor	North	B	other	6	387.85	0.03	No
17	2024-05-24	Angela Lindsey	South	C	agree	1	283.29	0	Yes
18	2021-06-07	Michael Scott DDS	North	A	discover	16	457.16	0.09	Yes
19	2024-03-06	Matthew Jenkins	North	C	seven	17	358.38	0.18	Yes
20	2023-10-12	Sarah Wall MD	South	C	Congress	16	456.18	0.27	No
21	2022-08-04	Christian Anderson	North	B	people	9	415.89	0.04	No
22	2024-04-01	Diane Harrell	East	B	free	6	341.92	0.27	No
23	2020-09-21	Manuel Wright	East	B	quickly	20	358.7	0.27	No
24	2022-02-27	Paul Franklin	North	B	because	11	463.91	0.11	Yes
25	2025-03-14	Cynthia Nelson	West	C	Republican	15	475.91	0.21	No
26	2021-05-04	Daniel Hooper	North	B	concern	12	221.01	0.25	No
27	2022-10-12	Karen Peterson	North	B	bill	5	313.15	0.02	Yes
28	2022-08-06	Melissa Moore	North	A	onto	6	24.9	0.14	Yes
29	2020-08-19	Christopher Montoya	East	B	us	12	286.18	0.3	No
30	2024-08-11	Courtney Young	East	B	often	2	158.62	0.2	Yes
31	2025-07-09	Michael Goodman	West	B	south	3	239.86	0.18	Yes
32	2023-02-14	Gary Young	North	C	walk	10	141.11	0.06	No
33	2020-04-13	Kristen Case	East	B	per	2	59.7	0.13	Yes
34	2021-10-26	Mark Carroll	East	B	during	16	408.32	0.29	Yes
35	2022-09-09	Holly Perez	North	C	represent	9	421.22	0.21	No
36	2022-11-19	Tiffany Hicks	East	C	drive	18	69.92	0.18	No
37	2020-08-04	Bradley Wallace	North	A	debate	2	466.05	0.04	No
38	2022-08-23	Heather West	East	A	seven	11	355.57	0.1	Yes
39	2020-10-28	Cynthia Rodriguez	East	B	have	9	220.17	0.07	Yes

40	2022-02-03	Tracy Wiggins	South	C	win	16	381.62	0.16	Yes
41	2025-01-27	Richard Collier	East	A	must	11	290.88	0.21	No
42	2021-04-20	Heather Gonzalez	West	C	outside	11	466.44	0.27	No
43	2024-02-29	Melissa Lewis	North	B	let	8	191.32	0.3	No
44	2022-06-01	Brian Clayton	West	A	eight	14	88.79	0.29	Yes
45	2022-04-07	Kelly Carr	North	B	program	5	240.54	0.22	Yes
46	2024-10-12	Brooke Stone	North	C	letter	9	172.73	0.19	Yes
47	2022-04-30	Sean Martin	West	C	sound	19	359.21	0.11	No
48	2021-09-07	Brenda Bishop	West	C	effect	10	416.89	0.29	Yes
49	2020-10-17	Angela Webb	South	B	who	1	183.34	0.06	Yes
50	2024-11-25	Jordan Tucker	East	C	meeting	7	88.37	0.03	No
51	2024-03-16	Katelyn Browning	West	C	training	10	420.35	0.28	No
52	2021-04-24	Sarah Bell	East	C	finally	20	243.26	0.09	No
53	2025-07-21	Mark Rodriguez	North	B	resource	19	485.26	0.01	No
54	2021-03-20	Cassie Munoz	South	B	past	19	299.96	0.29	Yes
55	2022-10-04	Tonya Harris	East	C	degree	7	39.65	0.05	No
56	2024-01-29	Jason Dickson	North	B	whole	6	75.41	0.28	Yes
57	2021-09-06	John James	South	B	night	19	479.95	0.08	No
58	2025-02-20	Joshua Parker	East	B	understand	7	321.65	0.11	No
59	2021-02-12	Patrick Contreras	West	C	summer	2	283.44	0.28	Yes
60	2024-01-18	Matthew Herrera	South	A	international	14	260.49	0.22	Yes
61	2024-04-19	April Campbell	East	A	whether	7	442.88	0.24	No
62	2022-03-16	Kristin Gordon	East	A	a	15	438.18	0.3	No
63	2024-04-29	Lisa Fitzgerald	North	C	area	13	345.39	0.25	No
64	2021-12-02	Paula Knox	South	A	reflect	1	113.04	0.01	Yes

65	2024-07-11	Carolyn Wood	North	A	garden	8	278.46	0.17	Yes
66	2022-01-23	Patricia Harvey	West	A	certain	5	418.9	0.29	Yes
67	2020-09-19	Christopher Douglas	East	C	care	8	481.75	0.2	Yes
68	2025-05-14	Cole Waller	North	B	current	16	167.19	0	Yes
69	2021-02-28	Timothy Howell	West	B	when	5	49.56	0.16	Yes
70	2022-03-21	Patricia Harris	West	A	talk	16	39.59	0.23	Yes
71	2022-08-31	Scott Hamilton	North	B	instead	1	347.8	0.05	No
72	2025-06-23	Nicole Morrison	West	A	thought	1	215.97	0.15	No
73	2024-08-11	Kelly Hill	West	B	community	19	105.11	0.15	No
74	2025-07-08	Beth Ramsey	West	A	determine	11	434.51	0.28	Yes
75	2021-12-16	Brenda Smith	East	B	federal	1	260.49	0.13	No
76	2023-04-30	Charlotte Patel	East	C	bring	13	351.45	0.24	No
77	2021-10-03	Chad Scott	East	B	floor	3	170.33	0.04	Yes
78	2022-08-08	Ryan Hammond	North	C	have	12	455.92	0.18	No
79	2023-06-07	Bridget Morris	South	B	young	2	181.96	0.09	No
80	2020-07-25	Samuel Hester	North	C	marriage	19	419.88	0.1	No
81	2023-12-27	Sarah Austin	West	B	enter	13	124.69	0.16	No
82	2023-01-31	Mr. Andrew Montgomery DVM	East	C	front	18	85.71	0.14	No
83	2022-10-09	Mrs. Kelly Ibarra	South	C	beat	8	189.74	0.11	No
84	2022-02-17	Nicholas Greene	South	C	ten	4	120.75	0.24	No
85	2021-07-30	James Moore	North	A	catch	2	105.62	0.3	Yes
86	2024-11-19	Edward Hinton	East	A	girl	15	466.63	0.24	Yes
87	2023-09-12	Alexa Avila	East	B	artist	10	125.06	0.01	Yes
88	2022-11-26	Kenneth Knapp	North	B	item	15	394.46	0.23	Yes

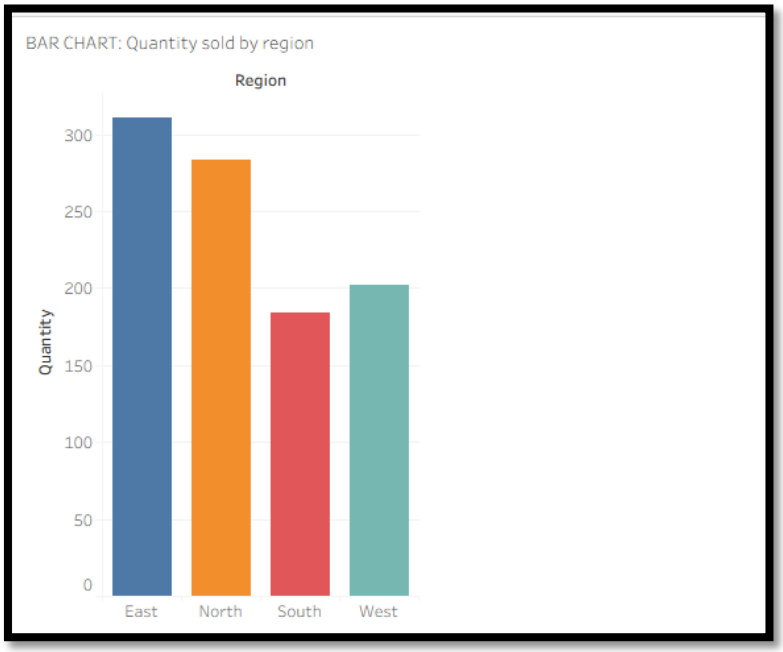
89	2022-02-12	Stacie Haney	South	C	lose	15	407.53	0.17	Yes
90	2023-02-27	Amanda King	East	B	my	11	273.86	0.27	Yes
91	2020-04-14	Linda Barr	East	C	lawyer	4	320.82	0.14	No
92	2020-06-28	Stephen Lane	East	B	look	4	264.21	0.29	Yes
93	2023-11-01	Barbara Kelly	North	B	score	11	423.33	0.21	No
94	2022-08-29	Nicole Frazier	West	B	society	6	239.14	0.26	Yes
95	2023-09-25	Michelle Gomez	South	A	these	17	121.8	0.17	Yes
96	2023-10-26	William Santiago	East	C	former	9	147.01	0.17	Yes
97	2020-10-05	Tracy Grimes	South	B	two	18	200.68	0.13	No
98	2020-04-26	Veronica Cruz	East	C	year	3	474.96	0.17	No
99	2020-09-10	Laura Simmons	South	C	challenge	10	42.55	0.06	Yes
100	2022-10-16	Anthony Eaton	West	C	open	18	77.5	0	Yes

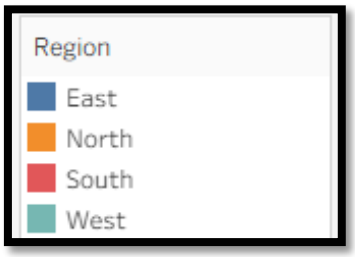
Table 3.1.1 – dataset of healthcare patient record

The dataset consists of 100 purchase records with following columns/ fields –

- **ID:** unique identifier for each patient
- **Date:** Date of patient visit or hospital admission
- **Customer:** name of the Patient
- **Region:** Location of hospital or patient origin
- **Category:** Grouping of medical condition or diagnosis
- **Product:** Specific diagnosis, condition, or procedure/treatment
- **Quantity** Number of visits, doses, or procedure count (depending on context)
- **Unit price:** Cost of a treatment, test, or visit
- **Discount:** Proportion of cost covered by insurance or subsidy
- **Returned:** Whether the patient returned (readmission, follow-up, or complication)

3.2 – Charts, Tables, and Interpretation ✞
BAR CHART –





3.2.1-sum of quantity sold by region

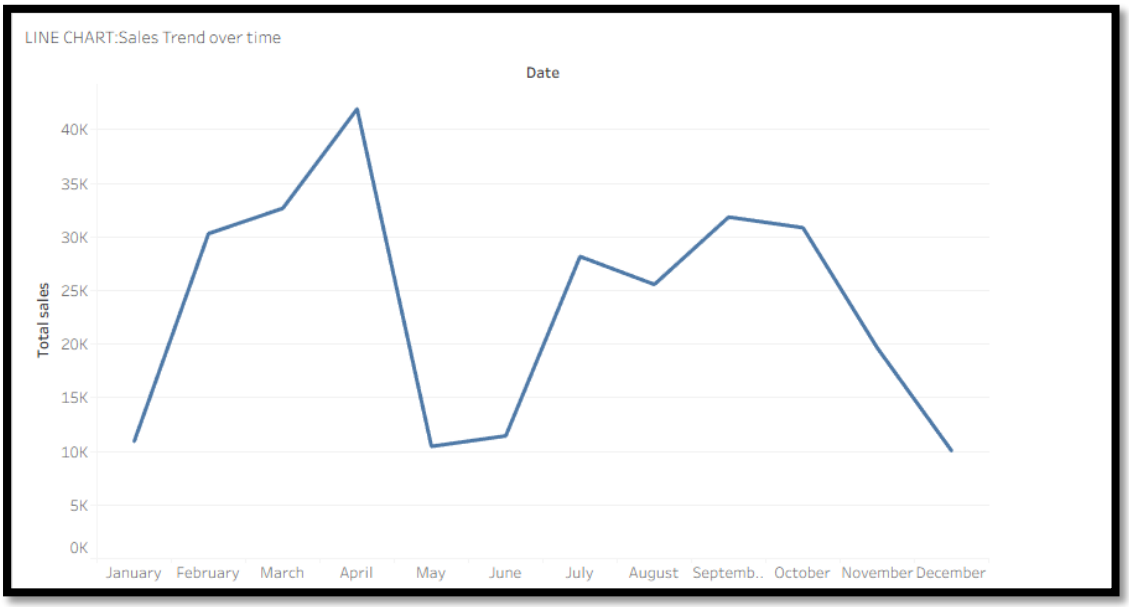
This bar chart displays the sum of quantity of products sold in four regions.

Region Interpretation

East	-	Highest number of patients treated or services provided
North	-	Second highest volume of patient records or treatments
West	-	Moderate patient volume
South	-	Lowest number of patients or records

.

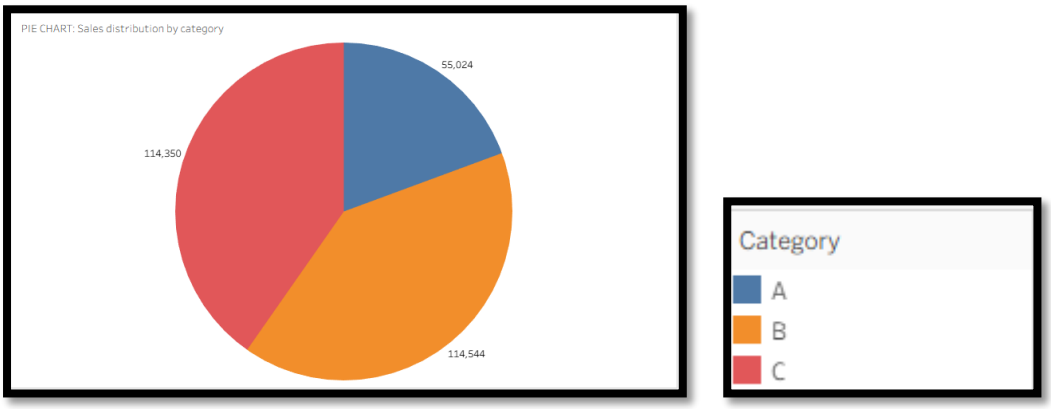
✞ LINE CHART –



3.2.2-sales trend over months

- **Seasonal Trends**:
  - April and September-October may correspond with **disease outbreaks** (e.g., flu season, respiratory illnesses).
  - May and December could be **holiday months** with reduced hospital visits or reporting.
- **Operational Factors**:
  - Drops could be due to **facility closures**, **staffing issues**, or **data reporting delays**.
  - Peaks could reflect **health campaigns**, **screenings**, or **policy changes** increasing patient intake.
- **Policy or Program Impact**:
  - A new public health program might explain spikes in certain months.
  - Insurance enrollment periods or subsidy changes could also affect patient numbers.

✞ PIE CHART –



3.2.3-sales distribution by category

Category B — Largest Share

This segment contributes the **highest number of patient records** among all categories. It represents the **biggest portion of the pie**, indicating it is the **dominant category** in terms of healthcare services or conditions recorded.

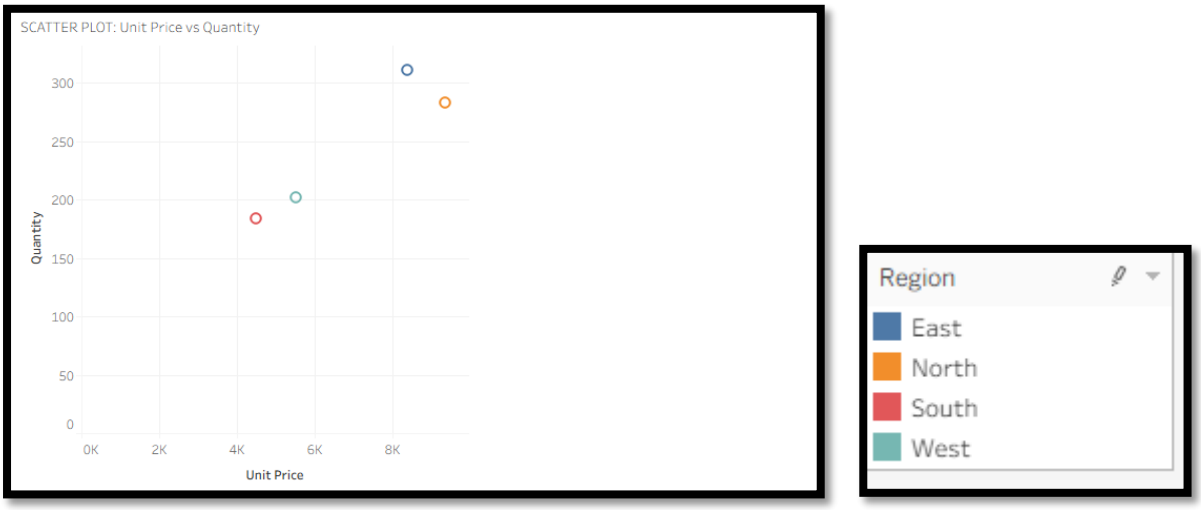
• Category C — Second Largest Share

This category holds a **moderate share** of patient records. It is **slightly smaller than Category B** but still represents a **significant portion** of the healthcare data, reflecting strong patient engagement in these services.

• Category A — Smallest Share

This is the **least contributing category** in the dataset. It represents the **smallest portion of patient records**, suggesting either **lower demand**, **underutilization**, or potential **gaps in data capture or service access** for this category.

SCATTER CHART-



3.2.4-unit price VS category

The scatter plot shows that some **high-cost treatments are also highly used**, indicating their importance in patient care. Mid-priced treatments have moderate usage, while some lower-cost services are less utilized. This suggests a need to review resource allocation and ensure cost-effective healthcare delivery.

TOTAL SALES

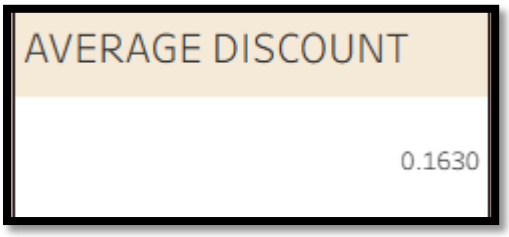
3.2.5-TOTAL SALES

The visual shown represents the **total cost of healthcare services** delivered over the selected period or across the complete dataset.

When we sum up the **cost of treatments, procedures, and consultations** provided to all patients, the cumulative amount reaches **283,918**.

This **KPI (Key Performance Indicator)** card is designed to provide a quick snapshot of **overall healthcare activity and expenditure**. Such a metric is crucial for **hospital administrators and healthcare decision-makers** because it reflects the **scale of services delivered**, and serves as a **benchmark** for measuring performance against **budgets, previous periods, or healthcare standards**.

- **AVERAGE DISCOUNT**



3.2.6- AVERAGE DISCOUNT

The visual highlights the **average discount or cost reduction** applied across all patient treatments in the dataset. The calculated value of **0.1630** indicates that, on average, patients received a **16.30% reduction in billed healthcare costs**.

This discount may come from:

- **Insurance coverage**
- **Government health schemes**
- **Hospital-provided financial aid or subsidies**

✞ RETURN RATE

RETURN RATE
1.000

Table 3.2.7 – Return Rate

A **100% return rate** reflects **strong patient retention** and **continuous care delivery**, which may indicate:

- Effective **follow-up scheduling systems**
- High levels of **patient satisfaction or trust**
- Management of **chronic or long-term conditions** requiring repeated care
- Successful implementation of **treatment plans** or **monitoring protocols**

✞ PROFIT

PROFIT
234,483

Table 3.2.8 – Profit

The healthcare facility generated a **net profit of 234,483**, indicating efficient cost management and revenue generation from patient services. This reflects strong financial performance and supports the sustainability of healthcare operations

4. FINDINGS & OBSERVATION

FOR TABLEAU:

4.1- Dashboard with tables and charts, outcomes

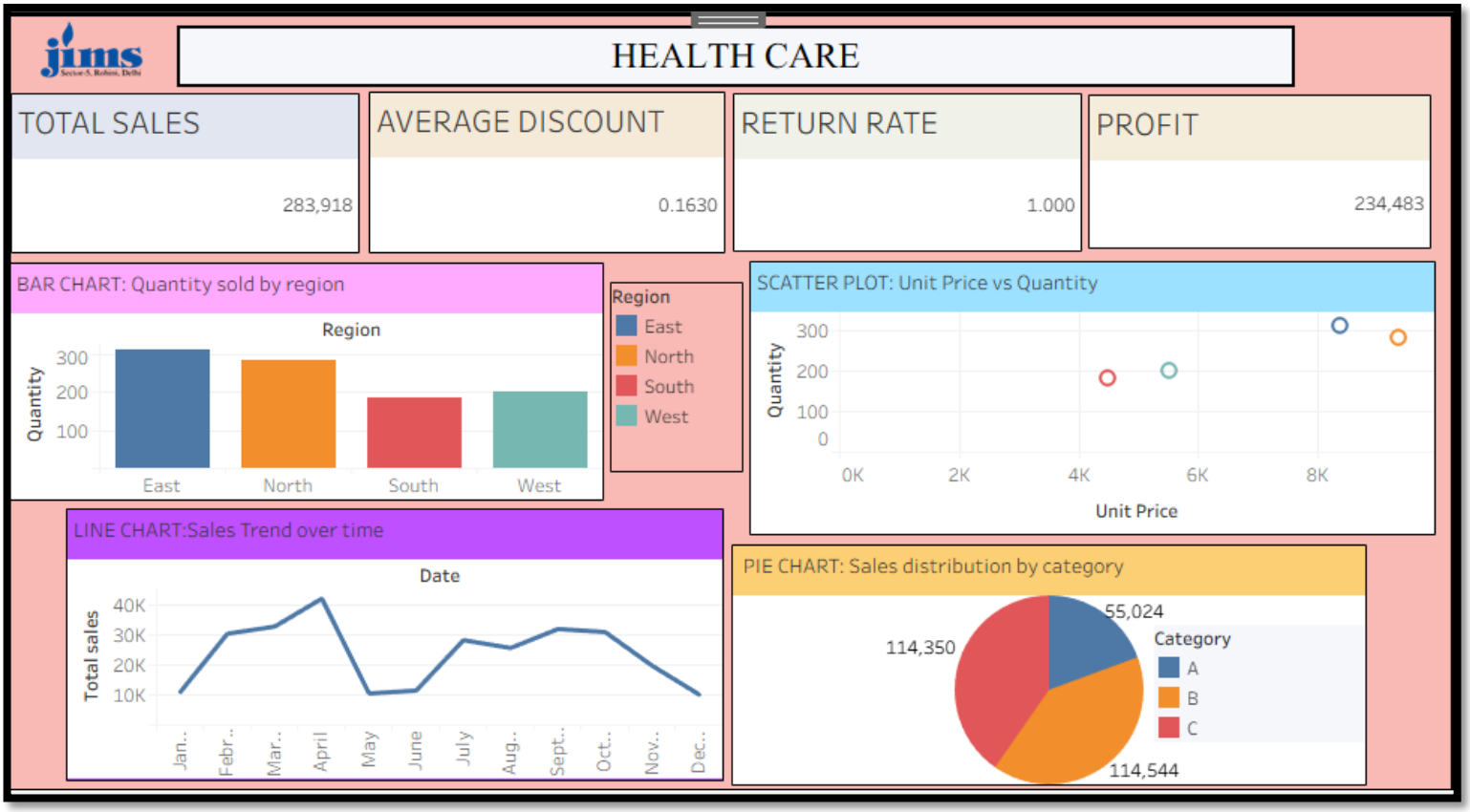


Figure 4.1.1 - Tableau dashboard for measures and charts

This dashboard of a healthcare patient record analysis presents four key measures:

- **Total sales** (283,918)
- **Average Discount (**0.1630)
- • **Return Rate** (1000)
- **Profit** (234483).

The dashboard includes multiple charts for better insights:

- **Quantity by Region (Bar Chart):**

- Highest treatment volume in the **East** and **North** regions.
- **South** and **West** show lower patient volumes.

- **Monthly Service Trends (Line Chart):**

- **April** shows the highest spike in service revenue.
- **May and December** show noticeable drops, suggesting seasonal fluctuations.

- **Service Category Distribution (Pie Chart):**

- **Category B** dominates in service volume (114,544), followed closely by **Category C** (114,350).
- **Category A** contributes the least (55,024), suggesting a narrower service scope or lower demand.

- **Unit Price vs Quantity (Scatter Plot):**

- High-cost services still see high usage, reflecting essential or critical care.

- Lower-priced treatments are moderately used.

Each chart and measures are interactive and navigates to its respective detailed sheet for deeper analysis. The JIMS logo at the top is hyperlinked to the official JIMS website.

Overall, the system demonstrates healthy growth, with certain regions and categories driving performance, while also indicating opportunities for strategic pricing and regional sales enhancement.

3. CONCLUSION AND RECOMMENDATION

5.1 - Summary Of Learning

The healthcare patient record analysis provided valuable insights into service delivery, patient behavior, and financial performance. Key findings include:

- **Total revenue** generated was significant (**283,918**), with a strong **profit margin** (**234,483**), indicating efficient operations.
- An **average discount rate of 16.30%** reflects balanced affordability for patients, likely through insurance or subsidy programs.
- A **100% return rate** suggests excellent patient engagement and follow-up compliance, though it may also indicate a high volume of chronic care cases.
- **Regional analysis** showed that the **East and North** regions had the highest patient volumes, highlighting disparities in regional service usage.
- **Sales trends** over time revealed seasonal fluctuations, with peaks in **April and September**, and lows in **May and December**.
- **Service category analysis** identified **Category B and C** as the most utilized, pointing to higher demand for certain types of treatments or specialties.
- **Unit price vs. quantity analysis** indicated that even **high-cost services** are widely accessed, suggesting their critical nature or urgent demand.

Recommendations:

1. **Improve Resource Allocation:**
   - Increase support in **low-performing regions** (South and West) to improve patient outreach and service access.
2. **Targeted Investment in High-Demand Services:**
   - Expand capacity and staffing in **Category B and C** areas to meet patient demand and prevent service delays.
3. **Monitor Chronic Care Workload:**
   - Given the high return rate, consider investing more in **preventive care** and **community-based health programs** to reduce hospital dependency.
4. **Evaluate Discount Strategies:**
   - Ensure that the average discount level remains sustainable while maintaining accessibility for all income groups.
5. **Plan for Seasonal Demand:**
   - Align staffing and supplies with identified **peak months** to ensure uninterrupted care and reduced wait times.
6. **Implement Data-Driven Decisions:**
   - Continue leveraging patient data dashboards to inform policy, operational improvements, and budgeting.

5.3 – Suggestion For Future Work

- Integrate clinical data such as diagnoses, treatment plans, and lab results for deeper analysis.
- Apply predictive analytics or machine learning to forecast patient trends, readmission risks, or resource demands.
- Segment data by age, gender, region, and other demographics to uncover disparities and tailor healthcare delivery.
- Develop real-time dashboards for monitoring patient flow, service usage, and emergency cases.
- Strengthen data privacy and security measures to ensure compliance with healthcare regulations.
- Include patient satisfaction and feedback data to evaluate service quality and improve patient experience.
- Conduct cost-versus-outcome analysis to assess the effectiveness and value of treatments.
- Incorporate staff performance data to evaluate efficiency and identify areas for professional development

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